

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 2 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|--------------------------------------|---|
| 1. Pivot B | (A) $A = LU$: a lower-triangular L times an upper-triangular U . |
| 2. Multiplier D | (B) The first nonzero entry in a row, used to clear the entries below it. |
| 3. Upper triangular (U) G | (C) A matrix with no inverse — its determinant $ad - bc$ is 0. |
| 4. Inverse matrix H | (D) $\ell = (\text{entry to clear}) \div (\text{pivot})$; how many pivot rows you subtract. |
| 5. Singular matrix C | (E) The identity with two rows swapped; it records a row exchange. |
| 6. LU factorization A | (F) The matrix A^T formed by flipping A across its diagonal (rows $\leftrightarrow$ columns). |
| 7. Permutation matrix E | (G) A matrix with all zeros below the main diagonal. |
| 8. Transpose F | (H) A^{-1} , the matrix that undoes A : $A^{-1}A = I$. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- To clear the entry below a pivot p , you subtract ℓ times the pivot row, where ℓ is (A) $p \times (\text{entry})$ (B) **(entry) $\div p$** (C) $p \div (\text{entry})$ (D) $p - (\text{entry})$
- In the factorization $A = LU$, the matrix L holds (A) the pivots (B) the solution $\mathbf{x}$ (C) **the multipliers, with 1's on the diagonal** (D) all zeros
- A 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is **singular** (no inverse) exactly when (A) $ad - bc = 0$ (B) $ad - bc = 1$ (C) $a = d$ (D) it is upper triangular
- If P is a permutation matrix, the product PA (A) scales the rows of A (B) **swaps rows of A** (C) transposes A (D) inverts A
- For any two matrices that can be multiplied, $(AB)^T =$ (A) $A^T B^T$ (B) AB (C) BA (D) **$B^T A^T$**
- Two equations in two unknowns have **infinitely many** solutions when the two lines (A) cross once (B) are parallel but different (C) **are the same line** (D) are perpendicular

Teacher Note

Item 2: L records the elimination multipliers (with unit diagonal); U holds the pivots. Item 5: socks-and-shoes order reverses — $(AB)^T = B^T A^T$, not $A^T B^T$. Item 6: same line = one equation is a multiple of the other (dependent) $\Rightarrow$ infinitely many; parallel-but-different $\Rightarrow$ none.

Part C — Short Answer & Computation (30 pts)

Show your work.

1. Solve the system by **elimination**, then back-substitute and check. State the pivot and the multiplier you used.

$$x + 2y = 4, \quad 3x + 4y = 10.$$

Pivot = 1 (the x -coefficient of eq. 1); multiplier $\ell = 3 \div 1 = 3$. Eq. 2 $- 3 \times$ eq. 1: $(4 - 6)y = 10 - 12$, so $-2y = -2$, $y = 1$. Back-substitute: $x + 2(1) = 4$, $x = 2$. Solution $(x, y) = (2, 1)$. Check: $3(2) + 4(1) = 10$ ✓.

2. Row-reduce $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 4 & 10 & 8 \end{bmatrix}$ to upper-triangular U . List the three multipliers and the pivots. $R_2 - 2R_1$ ($\ell_{21} = 2$) and $R_3 - 4R_1$ ($\ell_{31} = 4$) give $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 4 \end{bmatrix}$; then $R_3 - 2R_2$ ($\ell_{32} = 2$) gives $U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$. Multipliers $\ell_{21} = 2$, $\ell_{31} = 4$, $\ell_{32} = 2$; pivots 1, 1, 2.

3. Let $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$. Use the $ad - bc$ formula to find A^{-1} , then check that $A^{-1}A = I$. $ad - bc = 2 \cdot 2 - 1 \cdot 3 = 1$, so $A^{-1} = \frac{1}{1} \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$. Check: $\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ ✓.

4. Using the inverse from the previous problem, solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ (compute $\mathbf{x} = A^{-1}\mathbf{b}$). $\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 6-4 \\ -9+8 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. Check: $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ ✓.

5. Factor $A = \begin{bmatrix} 2 & 4 \\ 0 & 7 \end{bmatrix}$ as $A = LU$. Write L and U , then verify $LU = A$. **Pivot 2, multiplier $\ell = 6 \div 2 = 3$. $R_2 - 3R_1$: $\begin{bmatrix} 0 & 7-12 \end{bmatrix} = \begin{bmatrix} 0 & -5 \end{bmatrix}$, so $U = \begin{bmatrix} 2 & 4 \\ 0 & -5 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$. Check $LU = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 0 & -5 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 7 \end{bmatrix} = A$ ✓.**

6. Using the L and U from the previous problem, solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ in **two sweeps**: first $L\mathbf{c} = \mathbf{b}$ (forward), then $U\mathbf{x} = \mathbf{c}$ (back). $L\mathbf{c} = \mathbf{b}$: $c_1 = 2$; $3c_1 + c_2 = 1 \Rightarrow c_2 = 1 - 6 = -5$, so $\mathbf{c} = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$. $U\mathbf{x} = \mathbf{c}$: $-5x_2 = -5 \Rightarrow x_2 = 1$; $2x_1 + 4 = 2 \Rightarrow x_1 = -1$, so $\mathbf{x} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Check: $\begin{bmatrix} 2 & 4 \\ 0 & 7 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ ✓.

7. (a) The matrix $A = \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}$ has a 0 in the top-left pivot spot. Write the permutation matrix P that swaps the two rows, and give PA . (b) Write the transpose of $B = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ and state whether B is symmetric. (a) $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, and $PA = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$ (now a nonzero pivot sits in the corner). (b) $B^T = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} = B$, so B is symmetric.

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

1. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. (a) Compute $ad - bc$. Does A have an inverse? What does that say about solving $A\mathbf{x} = \mathbf{b}$? (b) How many solutions does $A\mathbf{x} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ have? (c) How many does $A\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ have? Justify (b) and (c) using elimination or the row/line picture. (a) $ad - bc = 1 \cdot 4 - 2 \cdot 2 = 0$, so A is *singular* — no inverse. Elimination hits a zero pivot ($R_2 - 2R_1$ wipes out row 2), so $A\mathbf{x} = \mathbf{b}$ has either no solution or infinitely many, never a unique one. (b) Apply $R_2 - 2R_1$ to $\mathbf{b}$: $6 - 2(3) = 0$, giving $0 = 0$ — a true statement, so infinitely many solutions (the two equations are the same line). (c) $R_2 - 2R_1$ on $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$: $1 - 2(1) = -1$, giving $0 = -1$ — impossible, so no solution (parallel but different lines).
2. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$. (a) Compute $A^T A$ and check that it is symmetric. (b) Explain why $A^T A$ is *always* symmetric for any matrix A , using the reversal rule $(AB)^T = B^T A^T$. (a) $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$, so $A^T A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 5 \\ 5 & 5 \end{bmatrix}$ — the off-diagonal entries match, so it is symmetric. (b) Take the transpose of $A^T A$ and apply the reversal rule: $(A^T A)^T = A^T (A^T)^T = A^T A$, since $(A^T)^T = A$. A matrix equal to its own transpose is symmetric, so $A^T A$ is always symmetric.

Scoring Part D (5 pts each). D1: 1 pt (a) $ad - bc = 0 \Rightarrow$ singular/no inverse; 2 pts (b) infinitely many with the $0 = 0$ justification; 2 pts (c) none with the $0 = -1$ justification. Full credit needs the *reason* (same line vs. parallel), not just a count. D2: 2 pts (a) correct $A^T A = \begin{bmatrix} 10 & 5 \\ 5 & 5 \end{bmatrix}$ and noting equal off-diagonals; 3 pts (b) the general argument $(A^T A)^T = A^T A$ via the reversal rule. Watch for students writing $(AB)^T = A^T B^T$ (wrong order).